

```
---
title: "DATA 607 Week 3A – Global Baseline"
author: "Patricio Romero"
date: "2026-09-19"
output: html_document
---
```

```
```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
```
```

Introduction

Recommendation systems use information about users' preferences to predict how they may evaluate items they have not yet rated. In this project, movie ratings from several critics will be used to estimate an unknown rating. The specific objective is to predict the rating that Param would give to *Pitch Perfect 2*.

The project uses PostgreSQL to store the observed movie ratings and R to retrieve, prepare, analyze, and model the data. The prediction will be calculated using the Global Baseline method demonstrated in the course materials.

Data Collection

The original data were provided in the `MovieRatings` worksheet of the `MovieRatings.xlsx` workbook. The worksheet contained a rating matrix in which each row represented a critic and each movie appeared in a separate column.

The dataset included the following six movies:

- Captain America
- Deadpool
- Frozen
- Jungle Book
- Pitch Perfect 2
- Star Wars Force

Numeric cells represented observed ratings. Blank cells represented movies that a critic had not rated. Param's rating for *Pitch Perfect 2* was identified as unknown in the accompanying problem statement.

The blank cells and the question mark were treated as missing observations. They were not converted to zero and were not replaced with percentile-based estimates because the purpose of this analysis is to estimate the unknown rating through the recommendation model.

Planned Approach

The analysis will follow these steps:

1. Convert the original wide rating matrix into a long-format dataset.
2. Retain only observed numeric ratings.
3. Store the prepared data in PostgreSQL.
4. Connect R to the PostgreSQL database.
5. Retrieve the rating records into R.
6. Validate the retrieved data and confirm that Param's rating for Pitch Perfect 2 is missing.
7. Calculate the global mean rating.
8. Calculate the user bias for Param.
9. Calculate the movie bias for *Pitch Perfect 2*.
10. Combine these components to obtain the Global Baseline estimate for Pitch Perfect 2.
11. Compare the calculated result with the professor's example.
12. Identify all movies that Param has not rated.
13. Calculate a Global Baseline prediction for each unrated movie.
14. Rank the candidate movies by predicted rating and recommend the movie with the highest predicted rating.

The Global Baseline prediction will use the following equation:

\$\$
 $\hat{r}_{u,i} = \mu + b_u + b_i$
\$\$
where μ is the global mean, b_u is the user bias, and b_i is the movie bias.

Data Preparation

The original worksheet was transformed from wide format into a long-format dataset with the following variables:

- `critic`: name of the person who provided the rating
- `movie`: name of the rated movie
- `rating`: observed numeric rating

Only numeric ratings were retained. Missing values were omitted from the long-format dataset, resulting in 61 observed critic-movie ratings.

The prepared records were saved as `movie\_ratings.csv` and imported into the PostgreSQL table `movie\_ratings`. The table contains a generated identifier named `rating\_id` in addition to the three analytical variables.

Before beginning the analysis in R, the imported data were checked in PostgreSQL. The validation confirmed that the table contained 61 observations and that no record existed for Param and *Pitch Perfect 2*. The global mean of approximately 3.9344 and Param's observed mean of 3.5 were also consistent with the professor's example.

Data Retrieval from PostgreSQL

The next stage retrieves the observed ratings directly from the PostgreSQL database into R. This keeps the database as the source of the stored observations while R performs the statistical calculations and produces the final report.

```
```{r load-packages, message=FALSE, warning=FALSE}
library(DBI)
library(RPostgres)

```{r database-connection, echo=FALSE, message=FALSE, warning=FALSE}
con <- dbConnect(
  RPostgres::Postgres(),
  host = "127.0.0.1",
  port = 55433,
  dbname = "data607_week3",
  user = "postgres",
  password = Sys.getenv("DATA607_DB_PASSWORD")
)

```{r retrieve-data}
ratings <- dbGetQuery(
 con,
 "
 SELECT critic, movie, rating
 FROM movie_ratings
 ORDER BY critic, movie
"
)

head(ratings)
```

### ## Data Validation

The retrieved data were validated before calculating the Global Baseline estimate.

```
```{r validate-data}
data.frame(
  observed_ratings = nrow(ratings),
  param_pitch_perfect_rows = sum(
```

```

    ratings$critic == "Param" &
    ratings$movie == "Pitch Perfect 2"
  )
)
```

```

### ## Global Mean Rating

The first component of the Global Baseline Estimate is the global mean, which represents the average of all observed movie ratings in the dataset.

```

```{r global-mean}
global_mean <- mean(ratings$rating, na.rm = TRUE)

global_mean
```

```

The global mean rating is approximately **3.9344**. This value represents the overall average rating across all critics and all observed movie ratings.

### ## Param's Mean Rating and User Bias

The next component is Param's user bias. First, the mean of Param's observed ratings is calculated. The unknown rating for *Pitch Perfect 2* is not included in this calculation.

```

```{r param-user-bias}
param_mean <- mean(
  ratings$rating[ratings$critic == "Param"],
  na.rm = TRUE
)

param_bias <- param_mean - global_mean

param_mean
param_bias
```

```

Param's mean rating is **3.5**. The user bias is calculated as Param's mean rating minus the global mean. A negative bias indicates that Param's observed ratings are, on average, lower than the overall rating average.

### ## Pitch Perfect 2 Movie Bias

The movie bias measures how the observed ratings for *Pitch Perfect 2* differ from the overall global mean.

```

```{r pitch-perfect-bias}
pitch_mean <- mean(
  ratings$rating[ratings$movie == "Pitch Perfect 2"],
  na.rm = TRUE
)

pitch_bias <- pitch_mean - global_mean

pitch_mean
pitch_bias
```

```

### ## Global Baseline Estimate

The Global Baseline Estimate combines the global mean, Param's user bias, and the movie bias for *Pitch Perfect 2*.

```

```{r global-baseline-estimate}
global_baseline_estimate <- global_mean + param_bias + pitch_bias

global_baseline_estimate
```

```

Using the Global Baseline method, the predicted rating that Param would give to \*Pitch Perfect 2\* is approximately **2.28**.

## ## Validation Against the Course Example

The calculated Global Baseline Estimate was compared with the result provided in the course spreadsheet.

```
`r validate-estimate`
professor_example <- 2.2798594847775178

data.frame(
 R_Estimate = global_baseline_estimate,
 Professor_Example = professor_example,
 Difference = abs(global_baseline_estimate - professor_example)
)`
```

The estimate calculated in R is approximately **2.28**, which is consistent with the Global Baseline result demonstrated in the course spreadsheet. This confirms that the R implementation reproduces the expected calculation.

## ## Recommendation for Param

The course example also requires evaluating all movies that Param has not rated and recommending the movie with the highest Global Baseline prediction.

```
`r candidate-movie-predictions`
all_movies <- sort(unique(ratings$movie))

param Rated_Movies <- unique(
 ratings$movie[ratings$critic == "Param"]
)

candidate_movies <- setdiff(
 all_movies,
 param Rated_Movies
)

movie_bias_table <- data.frame(
 movie = all_movies,
 movie_mean = sapply(
 all_movies,
 function(m) {
 mean(
 ratings$rating[ratings$movie == m],
 na.rm = TRUE
)
 }
)
)

movie_bias_table$movie_bias <-
 movie_bias_table$movie_mean - global_mean

candidate_predictions <- movie_bias_table[
 movie_bias_table$movie %in% candidate_movies,
 c("movie", "movie_bias")
]

candidate_predictions$predicted_rating <-
 global_mean +
 param_bias +
 candidate_predictions$movie_bias

candidate_predictions <- candidate_predictions[
 order(
 candidate_predictions$predicted_rating,
 decreasing = TRUE
)
]
```

```
),
]
```

```
candidate_predictions
\\\
```

Param has not rated \*Jungle Book\* or \*Pitch Perfect 2\*. The Global Baseline prediction is approximately \*\*3.47\*\* for \*Jungle Book\* and \*\*2.28\*\* for \*Pitch Perfect 2\*.

Because \*Jungle Book\* has the higher predicted rating, it is the recommended movie for Param among the movies he has not yet rated.

### ## Conclusion

The analysis successfully implemented the Global Baseline Estimate using movie ratings stored in PostgreSQL and analyzed in R. The overall mean rating was approximately \*\*3.9344\*\*, Param's mean observed rating was \*\*3.5\*\*, and his user bias was approximately \*\*-0.4344\*\*. \*Pitch Perfect 2\* had an average observed rating of approximately \*\*2.7143\*\* and a movie bias of approximately \*\*-1.2201\*\*.

Combining these components produced a predicted rating of approximately \*\*2.28\*\* for Param and \*Pitch Perfect 2\*. This result is consistent with the example provided in the course materials.

The analysis was then extended to all movies that Param had not rated. The Global Baseline predictions were approximately \*\*3.47\*\* for \*Jungle Book\* and \*\*2.28\*\* for \*Pitch Perfect 2\*. Because \*Jungle Book\* had the highest predicted rating among the unrated movies, it is the recommended movie for Param.

The implementation therefore reproduces the expected Global Baseline calculation and uses the same method to generate a recommendation from Param's unrated movies.

### ## Database Connection Cleanup

```
```{r disconnect-database}  
dbDisconnect(con)  
```
```

### ## AI Use

ChatGPT was used to help interpret the assignment requirements, organize the planned approach, improve the English writing, select an appropriate database structure, and provide coding guidance. I prepared and imported the movie-rating data, ran the code, reviewed the results, and confirmed the conclusions myself.